TREY LUPO

AI/ML Engineer

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SUMMARY

ML engineer with embedded finance domain expertise — shipping production ML and LLM systems, not notebooks. First technical owner inside a commercial real estate firm: production ML pipelines, streaming LLM integrations, and AI tooling used by the CFO's org. Statistics (USF) + Rust systems programming (FerroML, PyPI) — from model math to production API. MIT IDSS Data Science & Machine Learning Certificate.

EXPERIENCE

FP&A Analyst — Franklin Street

Tampa, FL · Sep 2025–Present

- Shipped a production **ML deal-closure pipeline** (AUC **0.874**, calibration error 4.8%) — ExtraTrees + XGBoost with MAPIE conformal intervals, walk-forward temporal validation, engineered leak-safe features, 51 automated tests — replacing analyst-scored CRM data with calibrated probability estimates.
- Built full-stack **AI financial dashboard** (React 19/TypeScript + FastAPI + Azure SQL) with **AI-generated variance narratives via Claude API (streaming LLM integration)**, consolidated P&L, drill-downs, three-tier caching — replacing a **132-page monthly report** and cutting variance analysis from **8+ hours to under 1 hour**.
- Built **Oracle NSPB / EPM Cloud automation** — production ETL for revenue and valuation-model uploads with **114+ tests** (open-sourced as the *nspb-api* SDK).
- Domain grounding: CRE analytics informing decisions across a **$500M+ portfolio**; surfaced a **$270B+** multifamily data gap that informed the CFO's board presentation.

AI Engineering Intern — Origin Construction

Jun 2025–Sep 2025

- Automated project-management workflows, integrated Procore APIs, and piloted AI agents for construction operations.

PROJECTS

- **FerroML** — statistically rigorous AutoML framework in **Rust** with Python bindings; **55+ algorithms**, **4,600+ Rust tests**; published on **PyPI**.
- **KKBOX Churn Prediction** — leak-safe point-in-time features (temporal-safety tested), isotonic calibration, temporal backtests, PSI drift monitoring on a **30GB+** user transaction dataset; **0.9696 AUC** on the tuned validation window, with no held-out test set; live on Hugging Face Spaces.
- **Local-LLM orchestration (Ornith)** — designed and operate a 24/7 system using Claude Code as orchestrator delegating implementation to locally served 9B/35B-MoE models (Ollama): propose-only task loop, sandboxed verification gates (bwrap, no-network), git-worktree isolation, human-review inbox; model-written code merges only after automated checks and review.
- **treylupo.com** — portfolio site with an embedded AI answer engine: agentic retrieval (RAG) over projects and resume with streaming, citation-grounded answers; Anthropic API, serverless SSE, rate-limited, eval-suite-verified. Full project set at treylupo.com.
- **nspb-api** — complete Python SDK for Oracle EPM Cloud (Planning, Migration, Data Management); **60+ API endpoints**.
- **Hierarchical Revenue Forecast** — open-source generalization of the production deal-closure work: multi-LOB reconciliation, SHAP explainability dashboard (Python/Go/TypeScript).
- **Open-Vocab Object Detection** — text-prompted detection + segmentation (Grounding DINO + SAM 2), 265–490 ms/image on RTX 3070.

EDUCATION

- B.A. Statistics, University of South Florida — May 2024
- Data Science & Machine Learning Certificate, MIT IDSS — 2024

SKILLS

Languages: Python · Rust · TypeScript · SQL · R

ML / Data: scikit-learn · XGBoost · LightGBM · PyTorch · Pandas · FastAPI · MAPIE · SHAP

AI tooling: Claude API / Anthropic SDK · Claude Code

Finance: Oracle EPM / NSPB Cloud · NetSuite · FP&A · financial modeling

Web / infra: React · GSAP · Vite · Azure SQL